DRILL 27

ADVANCED TACTICS

NFPA 1001 Level 2 (JPR 5.3.1, 5.3.2) / UL FSRI Research

COORDINATED FIREGROUND VENTILATION & FLOW PATH CONTROL

⏱ **Time:** 45–60 min | 🚒 **Apparatus:** Pumper / Engine | 🦺 **PPE:** Full Structural Turnout Gear & SCBA

👥 **Roles:** Drill Leader (Facilitator) | Safety Officer (Watcher) | Interior Attack Team & Exterior Vent Team

🎯 OBJECTIVE & RURAL FIREGROUND VALUE

Every firefighter must understand modern fire dynamics (UL/NIST research): how opening doors and windows creates a **Flow Path** feeding fresh oxygen to ventilation-controlled fires, how uncontrolled ventilation triggers rapid flashovers, and how to execute **Coordinated Ventilation** synchronized with water application (including exterior transitional attack).

📦 TOOLS & APPARATUS SETUP

- Apparatus pumper with charged 1.75" (45mm) attack line (smooth bore or combination nozzle).
- Ventilation tools: Halligan bar, flathead axe, 6-ft pike pole.
- Station bay doors, window prop, or simulated structure layout.
- Smoke machine or cardboard flow-path demo box (dollhouse prop).

⚠️ SAFETY WATCH-OUTS & STOP RULE

⚠️ **The Flashover Clock:** In modern homes filled with petroleum-based synthetic furnishings (plastics, poly foam), opening a door or breaking a window to a ventilation-limited fire accelerates flashover in as little as **90 to 120 seconds**.

⚠️ **Wind-Driven Hazards:** Never vent on the upwind side if crews are operating inside! Wind forces superheated gases directly onto interior firefighters.

🛑 **THE STOP RULE: Shout "STOP!" immediately if an exterior window or door is opened before the interior nozzle crew is in position with a charged, flowing line.**

PHASE 1: UNDERSTANDING THE FLOW PATH & MODERN FIRE BEHAVIOR (15 MINUTES) — SLOW DRY RUN

1. **Legacy vs Modern Fires:**
   - Legacy homes (natural wood/cotton): Fire was fuel-limited; took 20+ minutes to flashover.
   - Modern homes (plastics/synthetics): Rapidly consume available oxygen and become **ventilation-limited (oxygen-starved)** while producing massive volumes of unburned fuel gases.
2. **The Flow Path Definition:**
   - The space between an air intake (open door/window) and an exhaust opening (chimney, roof hole, or upper window).
   - **Bidirectional Flow Path:** Fresh oxygen rushes in low; black superheated smoke pushes out high (neutral plane).
   - **Unidirectional Flow Path:** Opening where air is strictly entering (intake) or strictly exhausting under high pressure (blowtorch exhaust).

PHASE 2: DOOR CONTROL & FLOW PATH ISOLATION (15 MINUTES) — BUDDY COACHING

1. **The Entry Door:** The front door is a massive oxygen intake.
2. **Door Control Drill:**
   - Forcible entry team pops the door.
   - **Immediate Action:** Pull the door almost completely shut again, leaving only a 2-inch crack while awaiting the charged hoseline.
   - Chock or hold the door closed to starve the fire of fresh air while the nozzle team flakes hose and dons facepieces.
   - When attack team is on air with line charged: open door, enter low, and control the door behind the hose to minimize incoming air.

PHASE 3: SYNCHRONIZED COORDINATED ATTACK (TRANSITIONAL ATTACK) (20 MINUTES) — TIMED / STRESS

1. **The Exterior "Transitional" Knockdown:**
   - Common in rural NB when arriving with 2–3 firefighters who cannot legally enter IDLH under WorkSafeNB 2-in/2-out rules.
   - Position at an exterior window showing heavy fire.
   - Direct a **solid or straight stream steeply upward against the ceiling** of the fire room.
   - Flow for **15 to 30 seconds** without moving the stream around.
   - Water coats the ceiling, turns to steam, absorbs heat, and dramatically knocks down the fire without pushing heat into adjacent rooms.
2. **Coordinating the Interior Advance with Ventilation:**
   - Exterior vent team stands by at the designated exhaust window with Halligan/pike pole.
   - **DO NOT BREAK THE WINDOW YET.**
   - Interior nozzle team enters, advances to the fire compartment, and begins cooling the ceiling gases with water.
   - Interior Team Leader calls over radio: *"Interior to Vent, we have water on the fire! Take the Side Bravo window!"*
   - Vent team clears the window completely (glass, sash, and curtains) to establish clean exhaust.

⚠️ COMMON PITFALLS & FIREGROUND HAZARDS

- ❌ *Ventilating before water is ready:* Breaking windows while the engine is still laying hose feeds fresh oxygen to the fire, causing rapid structural flashover.
- ❌ *Failing to clear the window sash:* Leaving window sashes or heavy blinds in place reduces ventilation efficiency by over 50%. Clean the entire window opening from top to bottom!
- ❌ *Whipping the nozzle in circles during transitional attack:* Rapidly swirling the nozzle creates high air entrainment, acting like a giant fan that pushes heat throughout the house. Hold the stream still against the ceiling!

🗣️ 5-MINUTE TAILGATE DEBRIEF

1. ***"What happens to an oxygen-starved fire when we open the front door?", "Why must the vent team wait for the radio call from the nozzle team before breaking glass?"***

✅ APPARATUS RESET CHECKLIST

- ☐ Pike poles and Halligan bars inspected and remounted.
- ☐ Broken glass or prop debris swept and disposed of.
- ☐ Hose drained, rolled, and repacked.
- ☐ Attendance roster signed.

DRILL LEADER SIGNATURE	SAFETY OFFICER SIGNATURE	DATE COMPLETED	TRAINING HOURS